Comprehensive Technical Analysis of the GRAPHIA Infrastructure and AI Ecosystem: Technology Stack and Large Language Model Integration

Introduction and Infrastructure Context of the GRAPHIA Initiative

The GRAPHIA project ("Knowledge Graphs, AI Services and Next Generation Instrumentation for R&D in Social Sciences and Humanities"), funded under the European Union's Horizon Europe program (Grant Agreement No. 101188018) with a total budget of €8.18 million spanning January 2025 to December 2027, aims to bridge the structural gap between Social Sciences and Humanities (SSH) data and modern computational analysis methods[1].
Historically, SSH data has been characterized by high fragmentation, heterogeneity, contextual dependence, and multi-linguality, limiting its machine readability and reducing its potential for interdisciplinary reuse[3]. Coordinated by the OPERAS consortium with key European Strategy Forum on Research Infrastructures (ESFRIs)—including EHRI, E-RIHS, GGP, and RESILIENCE—alongside academic institutions like the University of Bologna and the L3S Research Center, GRAPHIA establishes a unified digital ecosystem[1].
The primary output of the initiative is the first comprehensive Social Sciences and Humanities Knowledge Graph (SSH KG), uniting previously isolated archives, sociological survey datasets, bibliographic registries, and cultural heritage objects[3]. The technological foundation of the project relies on the integration of Open Science practices, semantic web standards, hybrid graph modeling, high-performance computing (HPC) environments, and specialized Large Language Models (LLMs)[3]. Examination of architectural documents and project outputs allows for a detailed reconstruction of the complete software stack, technical frameworks, and LLM implementations.

Knowledge Graph Architecture and Core Technology Stack

The platform's infrastructure avoids centralized data ingestion, favoring a distributed, federated knowledge graph model[8]. This model ensures semantic interconnectivity between autonomous graphs across various research institutions without forcing physical data migration into a central repository[8].
Data representation is built upon a hybrid graph architecture combining Resource Description Framework Graph (RDF) and Labelled Property Graph (LPG) technologies[3]. The RDF standard is employed for external data linking, ensuring FAIR (Findable, Accessible, Interoperable,

Reusable) compliance and providing a formal ontological schema[8]. Concurrently, LPG models are utilized within computational nodes to execute graph algorithms, analyze network topology, and aggregate entities at high speeds[8]. The GRAPHIA Ontology establishes a Common Data Model that tracks provenance metadata, semantic transformations, and mapping rules across disparate sources[8].

For continuous data acquisition, extraction, and enrichment, GRAPHIA employs a Data Acquisition Platform alongside the SSH Citation Index framework[8]. Systematic evaluations within the project demonstrated that standard STEM-oriented bibliographic processing tools fail when applied to SSH literature due to non-standard citation patterns, monograph references, and multilingual footnotes[8]. The developed framework automates the extraction, indexing, and semantic linking of citations from SSH resources, integrating specialized PDF parsing tooling developed by the OpenCitations team[1].

The system deployment architecture is divided into two operational environments[8]. Production web services, APIs, and microservices are deployed on OKD (the open-source distribution of Red Hat OpenShift)[8]. Computationally heavy workloads—including the fine-tuning, training, and execution of LLMs and large-scale graph analytics—are hosted in High-Performance Computing (HPC) environments[8]. Interoperability across components is governed by the Technical Interoperability Framework using standardized application programming interfaces (APIs)[8].

Architectural Layer	Applied Technologies and Standards	Infrastructural Role in GRAPHIA
Semantic Modeling	RDF, LPG, GRAPHIA Ontology (OWL/SKOS), SPARQL[8]	Hybrid storage and semantic integration across federated SSH graphs[8].
Citation Data Ingestion	SSH Citation Index Framework, OpenCitations PDF parsers[3]	Automated extraction, enrichment, and indexing of complex SSH references[3].
Compute & Hosting	OKD (OpenShift/Kubernetes), HPC clusters[8]	Cloud-native microservices hosting and HPC environments for AI workloads[8].
Interoperability Layer	Technical Interoperability Framework, REST/GraphQL APIs, Zenodo[8]	Standardized FAIR protocols and open publication of artifacts[8].

Ecosystem of Large Language Models: From

LLM4SSH Models to Agentic Systems

A core pillar of GRAPHIA's technical stack is a domain-tailored AI ecosystem that integrates open-weight models, fine-tuned architectures, and application-level LLM agents[8]. Relying exclusively on commercial, general-purpose LLM APIs off-the-shelf was deemed unviable due to the lack of specialized academic context in standard training sets and data privacy/sovereignty constraints[5].

The LLM4SSH Initiative

The flagship development in AI is **LLM4SSH**—a suite of open-source / open-weight large language models designed specifically for representing and reasoning over knowledge in the social sciences and humanities[3]. Architecturally, LLM4SSH functions as a domain-adapted counterpart to general conversational agents, tuned for historical contexts, academic terminology, and multilingual SSH corpora[3].

Key requirements for LLM4SSH include low inference latency, reduced operational costs, and local deployment options on private servers or institutional HPC centers[8]. Local hosting guarantees GDPR compliance when handling sensitive research materials, such as sociological survey entries, demographic data, or archival records related to violent conflicts[8]. LLM4SSH performs conceptual extraction from unstructured texts, translates non-structured narratives into structured graphs, and offers a natural language interface for researchers[4].

Conversational Interfaces and Graph Navigation: The Quagga Agentic System

To allow non-technical researchers to query the federated SSH KG without writing SPARQL queries manually, Odoma developed the **Quagga** ecosystem (Question Answering Over Graphs) for GRAPHIA[8].

Built around Knowledge Graph Question Answering (KGQA) and Retrieval-Augmented Generation (RAG), Quagga consists of three interconnected modules[8]:

1. **Quagga Agent:** An agentic system built on open-weight LLMs using RAG[8]. The agent accepts natural language questions, extracts semantic context, and dynamically constructs executable SPARQL queries against the SSH KG and linked datasets (e.g., the GoTriple graph)[8].
2. **EXYGEN API (Explore Your Graphs Engine):** A service that extracts schema metadata from target SPARQL endpoints to ground the LLM[8]. EXYGEN feeds up-to-date class structures and predicates to the agent, minimizing syntax errors and hallucinations during query generation[8].
3. **Quagga Benchmark:** A crowdsourced evaluation dataset pairing natural language questions with verified SPARQL queries[8]. Containing over 250 verified pairs across 28 SSH knowledge graphs, this benchmark provides continuous fine-tuning and validation for the Quagga Agent[9].

Applied AI Stack in Specialized Pilot Use Cases

The application of LLMs within GRAPHIA extends to real-world research and industrial pilot implementations, testing varied neural network architectures and prompt-engineering strategies[8].

Inductive Thematic Analysis in TALLMesh

Developed by Abertay University, TALLMesh is a Python- and Streamlit-based web application for automated qualitative thematic analysis adhering to the Braun & Clarke framework[11]. The tool connects user interfaces to various LLM APIs, supporting commercial offerings (OpenAI GPT-4o, Anthropic Claude, Azure OpenAI) and open-weight models (Llama-3, DeepSeek, Google Gemini)[12]. Prompt optimization and reasoning chains within TALLMesh are managed via the **DSPy** framework, while code selection accuracy is evaluated using the Inductive Thematic Saturation (ITS) metric[14].

Automated Indexing in the EHRI Pilot

The European Holocaust Research Infrastructure (EHRI) pilot investigates automated subject indexing across heterogeneous, multilingual archival records[5]. Initial experiments evaluated traditional classification approaches against LLMs like GPT-4o, Claude, and Mistral[15]. To maintain archival precision, future integration relies on the local LLM4SSH infrastructure operating within human-in-the-loop workflows[15].

Impact Documentation in the IMeTo System

Jointly developed by IBL-PAN and PSNC, the IMeTo pilot fine-tunes LLMs on Poland's national RADON research database[11]. The model analyzes scientific publications to extract, categorize, and document the societal and economic impact of academic research outputs[6].

Spatial Analysis in Genius Loci

The industrial pilot Genius Loci (powering the Genius Property platform) uses LLM extraction models to link historical narratives and academic research from the SSH KG with cadastral, geospatial, and real estate data, generating dynamic indicators of regional attractiveness and cultural context[6].

Fundamental Social Simulation Models

In theoretical work linked to GRAPHIA, researchers employ post-training techniques for LLMs using Reinforcement Learning (RL) with structural rewards derived from Graph Neural Networks (GNNs)[10]. This approach optimizes LLM agents for micro- and macro-level social network simulations.

Tool / Pilot	AI Architecture / Models	Integration Mechanism	Research Function
LLM4SSH (Core)	Open-source domain-adapted LLMs[8]	Local inference on HPC & OKD[8]	Textual interface to SSH KG, domain-specific analytics, entity extraction[8].

Quagga System	Open-weight LLMs + RAG (Odoma)[8]	EXYGEN API + SPARQL endpoints[8]	Natural language translation to SPARQL queries for graph traversal[8].
TALLMesh	GPT-4o, Claude, Llama-3, DeepSeek, Gemini[7]	Streamlit GUI, REST APIs, DSPy framework[12]	Automated thematic coding and qualitative analysis of interview transcripts[6].
EHRI Pilot	Mistral, Claude, GPT-4o → transition to LLM4SSH[15]	Human-in-the-loop pipelines[15]	Multilingual subject indexing of archival collections[15].
IMeTo Pilot	Fine-tuned LLMs (RADON corpus)[6]	Plugin / Standalone microservices[6]	Extraction and scoring of societal research impact indicators[6].
Genius Property	Specialized LLM extractors[6]	Semantic extraction from SSH KG[6]	Linking qualitative SSH knowledge to geospatial real estate data[6].

Synthesis and Technical Outlook

GRAPHIA's architectural strategy reflects a systematic integration of modern AI techniques into humanities data processing. Rather than relying on simple API wrappers around commercial models, the project builds an end-to-end framework that directly targets SSH data isolation[4]. The core technical achievement lies in unifying formal semantics (RDF/LPG knowledge graphs) with generative AI via RAG and agentic query engines like Quagga[8]. By developing open-weight models under the LLM4SSH banner and deploying them on sovereign OKD/HPC infrastructure, GRAPHIA preserves data privacy, GDPR compliance, and research transparency while modernizing European digital infrastructures for the social sciences and humanities[3].

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